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Dear Editor,

We are pleased to submit our manuscript, *FloodLite: A Multi-Platform Edge-Deployment Benchmark for Lightweight Flood Segmentation*, for consideration as a Research Article in the *International Journal of Disaster Risk Reduction*.

Operational flood-extent mapping is bottlenecked not by segmentation accuracy on academic benchmarks — that problem is essentially solved at $\text{IoU} \approx 0.93$ on the public Flood Semantic Segmentation Dataset (FSSD) by current encoder–decoder architectures — but by the gap between published models and the constrained hardware that humanitarian responders actually deploy. Across the recent flood-segmentation literature, every paper reports IoU on a benchmark; **none reports per-platform inference latency, post-quantization accuracy, or model size**. Our manuscript closes that gap with the first systematic edge-deployment benchmark for flood segmentation, evaluated across two execution providers (Apple Silicon CPU and browser WebAssembly), three lightweight UNet students, and FP32 + INT8 precisions.

Two empirical findings make this work distinctive:

1. A negative result on knowledge distillation. Across three architectures $\times$ four distillation configurations $\times$ three folds, no configuration of response-, feature-, or combined-KD improves over the no-KD baseline; the best student (MobileViT-XXS-UNet, no-KD, FP32) *exceeds* the UNet+EfficientNet-B0 teacher by 0.011 IoU points. We trace this to the saturation of FSSD (663 images, $\text{IoU} \approx 0.93$) combined with strong ImageNet pretraining, which leaves no headroom for the teacher’s soft targets to transfer. We report this honestly rather than search for a configuration that would have justified a more conventional positive-result framing, because we believe the field is better served by an explicit benchmark-saturation caveat than by yet another marginal-gain KD recipe.

2. Architecture-dependent post-training quantization stability. Under an identical ONNX static QDQ INT8 recipe, EfficientNet-Lite0 quantizes cleanly ($\Delta \text{IoU} = -0.043 \pm 0.016$) while MobileNetV3-Small and MobileViT-XXS suffer 0.45-IoU collapses with high cross-fold variance. We trace the failures to the HardSwish activation family and to LayerNorm-plus-attention blocks, and we make a concrete recommendation: ReLU6-bounded backbones (such as EfficientNet-Lite0) are the safe default for INT8 PTQ on flood-segmentation pixels, and HardSwish- or attention-based backbones require quantization-aware training. To our knowledge this is the first systematic characterisation of this failure mode in the flood-segmentation literature, and it directly informs deployment-architecture choice for practitioners.

Our recommended deployable configuration — EfficientNet-Lite0-UNet INT8 served by ONNX Runtime CPU — runs at **58 FPS on a 2023 laptop CPU** and **6 FPS in a single-threaded WebAssembly browser sandbox** at **3.83× size reduction** and **95% IoU recovery vs. the teacher**. The same `.onnx` artifact serves both deployment targets, which we offer as a practical pattern for heterogeneous flood-response edge hardware.

The work fits IJDRR’s scope because the contribution is not a new neural-network design but a *deployment-readiness benchmark for disaster-response practice*: the metrics we report (per-platform latency, INT8 size, INT8 IoU stability) are the metrics a humanitarian-data-science team needs to make a deployment decision, and they have been systematically absent from the prior flood-segmentation literature. We hope this orientation is useful to IJDRR’s interdisciplinary readership of disaster-management practitioners and applied researchers.

Reproducibility and openness. All code, ONNX exports, per-fold checkpoints, and benchmarking harnesses are released under MIT licence at <https://github.com/m-ibrahim-khalil/flood-segmentation>. A Zenodo-archived snapshot (DOI to be minted at acceptance) will be linked in the camera-ready. The FSSD dataset is publicly available on Kaggle.

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We would be grateful for your consideration of this submission and welcome any feedback from the editorial team and reviewers.

Sincerely,

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On behalf of all authors